

IPW3 : NACA0012 Code-to-Code Comparisons

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September 21, 2026



**POLYTECHNIQUE
MONTREAL**

TECHNOLOGICAL
UNIVERSITY



Table of Contents

- ① Case Conditions & Submission Summary
- ② Aerodynamic Grid Convergence
- ③ Droplet Impingement
- ④ Convective Heat Transfer & Thermodynamics
- ⑤ Ice Accretion
- ⑥ NACA0012 Overall Assessment



Geometry and Aerodynamic Conditions



NACA0012 – Wind-Tunnel View

Geometry

Parameter	Value
Airfoil	NACA0012
Chord, c [m]	0.5334
Angle of attack, α [°]	4.0

Aerodynamic Conditions

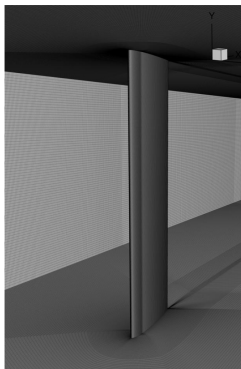
Parameter	Value
M_∞ [-]	0.3114
U_∞ [m/s]	102.15
T_∞ [K]	267.75
ρ_∞ [kg/m ³]	1.1884
Re_c [-]	3.8441×10^6

Quantities of Interest

$$C_L, \quad C_D, \quad C_M, \quad C_p$$



Icing Conditions



NACA0012 – Mesh Tunnel View

Grid	Nodes	Elements
L1	60.45 M	59.86 M
L2	27.31 M	26.99 M
L3	11.23 M	11.08 M
L4	5.95 M	5.86 M

	AE3932	AE3933
T_0 [°C]	−5.6	−2.2
T_∞ [K]	262.305	265.8
M_∞	0.3151	0.3127
U_∞ [m/s]	102.31	102.19
LWC [g/m ³]	0.55	0.55
MVD [μm]	20	20
Exposure time [min]	7	7
Exp. ice mass [g]	101.0	85.9

Experimental Observation

The warmer AE3933 condition produces approximately **15% less ice mass** than AE3932 (85.9 g vs. 101.0 g).

Droplet Distribution

- 1-, 3-, 7-, and 15-bin distributions
- Total LWC preserved across distributions
- Evaluated across all four grid levels

Quantities of Interest

$$\beta, \quad HTC, \quad T_s, \quad FF, \quad m_{\text{water}}, \quad m_{\text{ice}}, \quad m_{\text{evap}}$$



NACA0012 – Modeling Choices Across Submissions

- **Turbulence:** $k-\omega$ formulations were most common (**62.5%**), followed by SA (**37.5%**).
- **Droplet formulation:** Predominantly Eulerian (**75%**), with **25%** Lagrangian.
- **Heat transfer:** Documented approaches included recovery-temperature and boundary-layer integral methods.
- **Thermodynamics:** Messenger-based models were dominant (**71.4%**); SWIM-based and fluid-film approaches were also used.
- **Surface roughness:** Documented submissions used constant roughness heights ($k_s \approx 0.5$ mm).
- **Geometry evolution:** Most commonly based on Lagrangian/IB/node displacement; level-set, ALE morphing, and interpolation-based approaches were also reported.
- **Ice accretion strategy:** Both single-shot (**5 submissions**) and multilayer/multishot (**4 submissions**) approaches were represented.



NACA0012 – Summary of Submitted Data

AE3932

ID	L1				L2				L3				L4			
	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓				✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								✓
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008		*				*				*				*		
009		*				*				*				*		
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020				✓				✓	✓	✓	✓	✓				✓
Total	6	5	5	9	7	6	6	10	7	6	6	8	6	5	5	8

AE3933

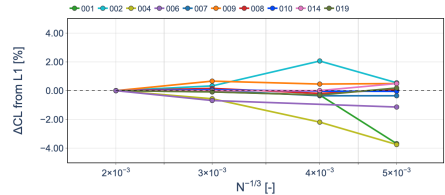
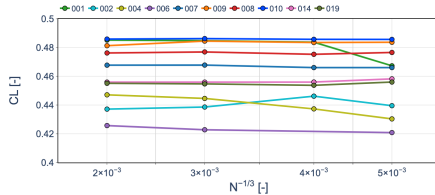
ID	L1				L2				L3				L4			
	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓	✓	✓	✓	✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
009	✓				✓				✓				✓			
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020												✓				
Total	8	6	6	8	8	6	6	8	8	6	6	7	7	5	5	6

✓: populated CutData received for the corresponding droplet-bin distribution.

*: Excel results only, no CutData slices.



NACA0012 – Lift Coefficient Grid Convergence

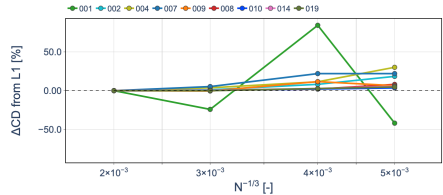
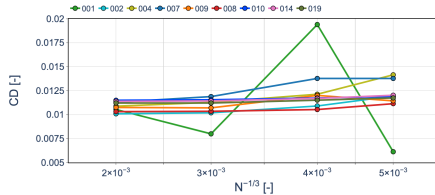


Observations

- At L1, the median lift coefficient is $C_L = 0.4677$, with an IQR of 0.0262 (5.6%).
- Most submissions exhibit weak sensitivity to spatial grid refinement.
- Relative to L1, coarse-grid differences remain within approximately 4% across the submissions.



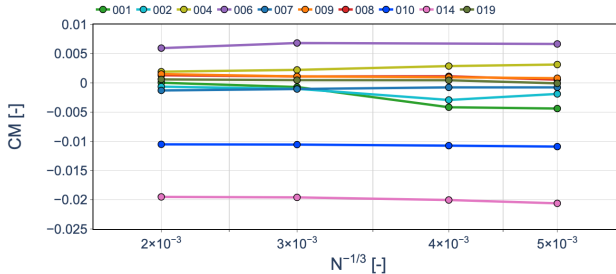
NACA0012 – Drag Coefficient Grid Convergence



Observations

- At L1, the median drag coefficient is $C_D = 0.01088$ (**108.8 drag counts**), with an IQR of **7.57 drag counts** (7.0%).
- C_D exhibits the greatest sensitivity to spatial grid refinement among the integrated aerodynamic coefficients.
- On the coarser grids, most submissions deviate by up to approximately **30%** from their respective L1 values.
- One submission additionally exhibits pronounced non-monotonic behavior with grid refinement.

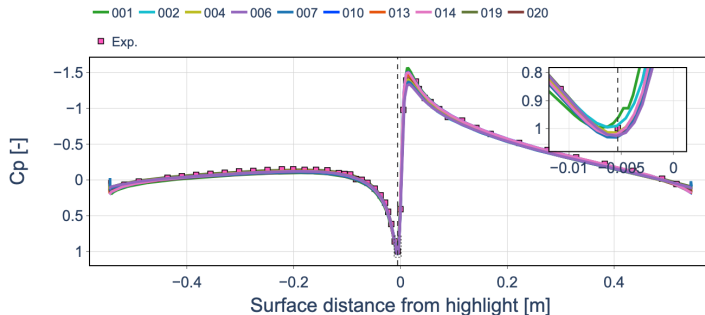
NACA0012 – Pitching Moment Grid Convergence



Observations

- At L1, the median pitching-moment coefficient is $C_M \approx 0$, with an IQR of 2.60×10^{-3} .
- Most submissions exhibit little variation in C_M across the grid hierarchy.

NACA0012 – Surface Pressure Distribution (L1)



Observations

- The L1 C_p distributions are tightly clustered and show good overall agreement with the experimental measurements.
- The largest code-to-code differences are localized near the leading-edge suction peak and stagnation region.
- Away from the leading edge, the predicted pressure distributions are nearly coincident across submissions.

NACA0012 – Aerodynamic Assessment Summary

L1 Code-to-Code Variability

	Median	IQR	Relative IQR
C_L	0.4677	0.0262	5.6%
C_D	108.8 counts	7.57 counts	7.0%
C_M	≈ 0	2.60×10^{-3}	–

Relative IQR is not reported for C_M because its median is close to zero.

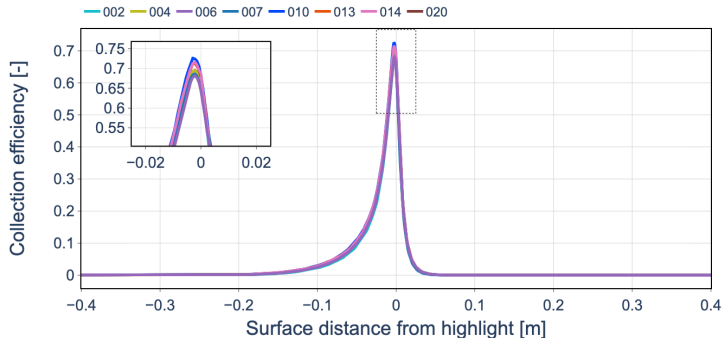
Grid-Refinement Behavior

- C_L and C_M exhibit limited within-code sensitivity to grid refinement for most submissions.
- C_D shows greater and less uniform grid sensitivity, including non-monotonic behavior for selected submissions.
- For most submissions, code-to-code differences exceed the changes produced by spatial refinement.

Aerodynamic Takeaway

Overall, the aerodynamic predictions are well clustered, with L1 C_p distributions showing good agreement with experiment despite differences in prescribed surface roughness. Among the integrated coefficients, C_D exhibits the greatest sensitivity to spatial grid refinement.

NACA0012 – Collection Efficiency (L1, 15 Bins)

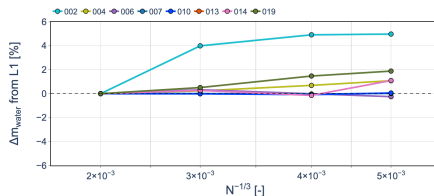
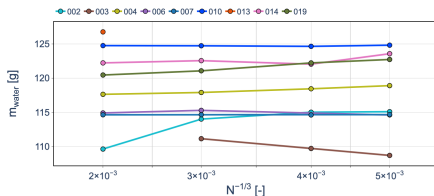


Observations

- The L1 collection-efficiency profiles are tightly clustered across submissions and exhibit very similar leading-edge behavior.
- Differences are concentrated near the peak region and impingement limits.



NACA0012 – Water Mass Grid Convergence (15 Bins)

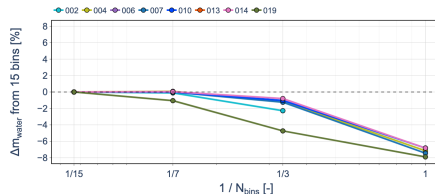
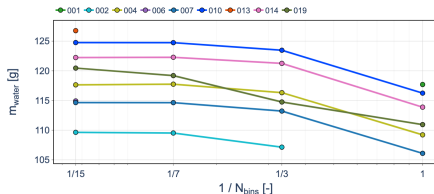


Observations

- For the 15-bin distribution, the L1 median impinging water mass is 119.05 g, with an IQR of 6.39 g (5.37%).
- Most submissions exhibit weak grid sensitivity and remain close to their L1 prediction across the grid hierarchy.



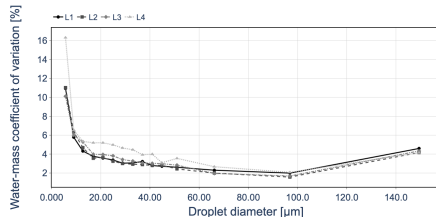
NACA0012 – Droplet Distribution Sensitivity (L1)



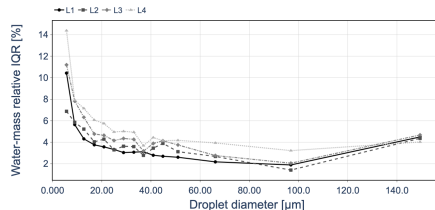
Observations

- The 15-bin and 7-bin distributions produce similar total impinging water masses.
- Reducing the distribution from 7 to 3 bins introduces only a modest additional change for most submissions.
- Relative to the 15-bin distribution, the single-bin approximation reduces the collected water mass by approximately 5–8% for most participants.
- Droplet-size discretization therefore introduces a clearer systematic effect than spatial grid refinement.

NACA0012 – Water-Mass Variability by Droplet Diameter (15 Bins)



Coefficient of Variation

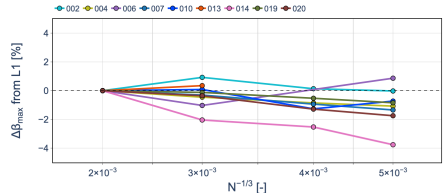
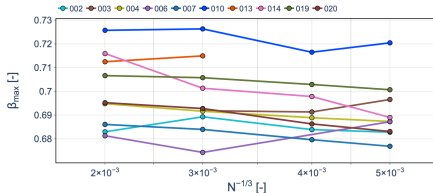


Relative IQR

Observations

- Code-to-code variability is strongly dependent on droplet diameter, with the largest spread observed for the smallest droplets.
- Variability decreases rapidly with increasing diameter and remains comparatively low over most of the intermediate droplet-size range.
- The coarser L3–L4 grids generally exhibit greater variability than L1–L2, particularly for the smallest droplets.
- A slight increase in variability is observed again for the largest droplets, potentially reflecting greater sensitivity to differences in droplet-drag treatment and SLD modeling assumptions.

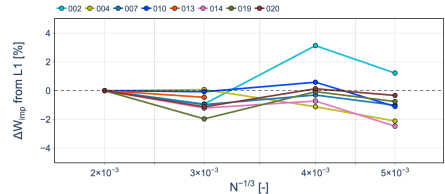
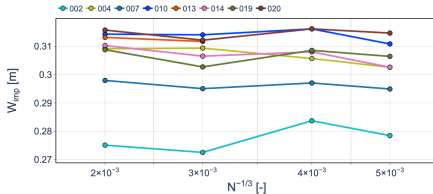
NACA0012 – Peak Collection Efficiency Grid Convergence (15 Bins)



Observations

- At L1, the median peak collection efficiency is $\beta_{\max} = 0.695$, with an IQR of 0.0209 (3.0%).
- Most submissions exhibit limited sensitivity of $\beta_{\max}$ to spatial grid refinement.
- Similar code-to-code clustering and grid-refinement behavior are observed for the other droplet-size distributions.
- The location of $\beta_{\max}$ was also examined; however, it is strongly influenced by the local leading-edge surface discretization. Consequently, no clear grid-convergence conclusion is drawn for the peak position.

NACA0012 – Impingement Width Grid Convergence (15 Bins)



Observations

- At L1, the median impingement width is 0.309 m, with an IQR of 8.93×10^{-3} m (2.9%).
- The impingement width is not fully grid-converged, although the variations between successive grid levels remain relatively small for most submissions.
- The remaining grid dependence partly reflect the sensitivity of the impingement limits to the surface discretization.



NACA0012 – Droplet Impingement Assessment

Spatial Grid Convergence

- Water mass, $\beta_{\max}$, and impingement width show limited grid sensitivity for most submissions.
- Code-to-code variability is generally comparable to or larger than the grid-refinement effect.
- Peak position was examined, but its sensitivity to the local leading-edge discretization prevented a clear convergence assessment.

Droplet-Size Effects

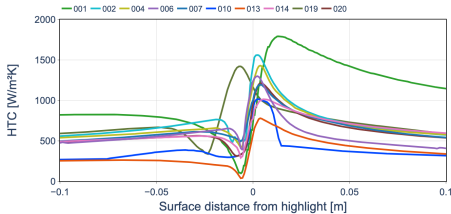
- The 7-bin distribution closely reproduces the 15-bin results, while the single-bin approximation systematically underpredicts collected water mass.
- Code-to-code variability is largest for the smallest droplets and decreases substantially over the intermediate diameter range.

Impingement Takeaway

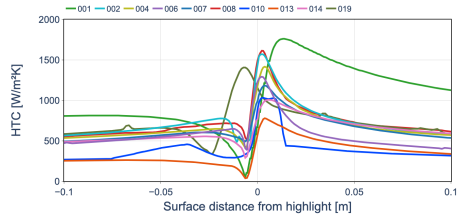
Overall, the impingement predictions are relatively consistent with spatial grid refinement. Droplet-size discretization has a clearer systematic influence, with 7 bins providing results close to the 15-bin reference.



NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932



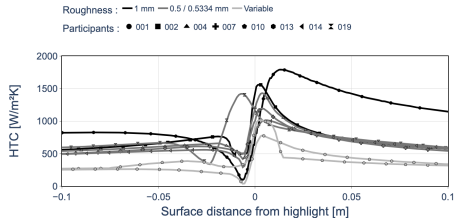
AE3933

Observations

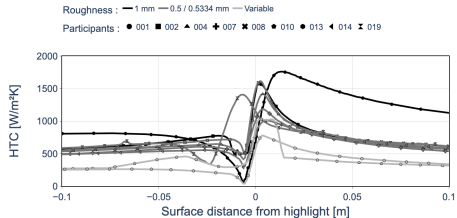
- Considerable code-to-code variability is observed in *HTC*, particularly near the leading edge, with similar behavior for both icing conditions.
- Prescribed surface roughness contributes systematically to the spread, but does not fully explain the observed code-to-code differences.



NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932 – Grouped by Roughness



AE3933 – Grouped by Roughness

Observations

- Considerable code-to-code variability is observed in HTC , particularly near the leading edge, with similar behavior for both icing conditions.
- Prescribed surface roughness contributes systematically to the spread, but does not fully explain the observed code-to-code differences.



NACA0012 – Integrated Convective Heat Transfer

Grid-Convergence Metric

To assess whether the differences in the local HTC distributions translate into a clear grid-convergence behavior, an integrated convective heat-transfer quantity is evaluated along the surface slice:

$$Q'_c = \int HTC(s) [T_s(s) - T_{rec}(s)] ds \quad [Q'_c] = \text{W/m}.$$

The submitted HTC and surface temperature T_s are used directly. The recovery temperature T_{rec} , however, was not part of the submitted dataset and must therefore be reconstructed.

Recovery Temperature Reconstruction

The submitted pressure coefficient is used to estimate the local external-flow state and subsequently the recovery temperature:

$$C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{rec}(s)$$

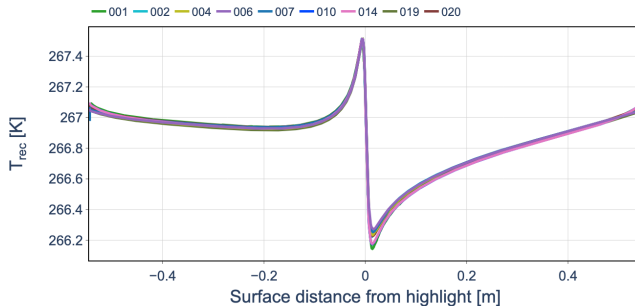
Assuming $p_s \simeq p_e$ and an isentropic external flow,

$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_{p,e},$$

with

$$T_{rec} = T_e \left[1 + r \frac{\gamma - 1}{2} M_e^2 \right], \quad r = Pr^{1/3} \approx 0.9.$$

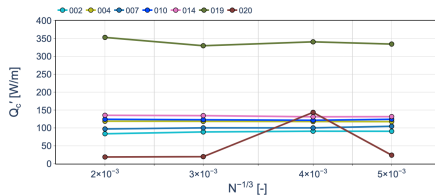
NACA0012 – Reconstructed Recovery Temperature (L1)



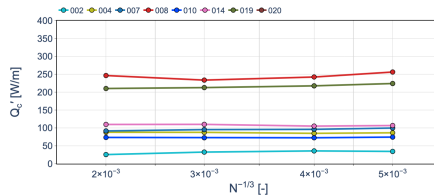
Observations

- The reconstructed T_{rec} distributions are closely clustered across submissions.
- This consistency follows directly from the close agreement previously observed in the submitted C_p distributions.
- Differences in Q'_c are therefore expected to originate primarily from HTC and T_s , rather than from the reconstructed T_{rec} .

NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932



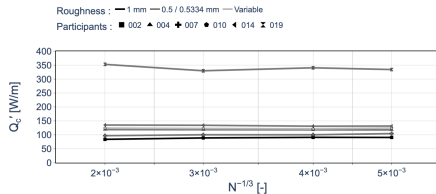
AE3933

Observations

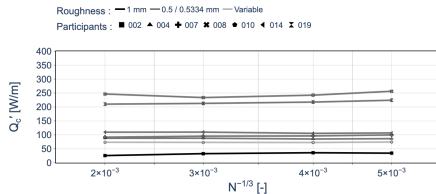
- Most submissions exhibit limited grid sensitivity in the integrated Q'_c .
- Grouping by prescribed roughness reveals a systematic roughness effect, although significant code-to-code variability remains within the roughness groups.



NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932 – Grouped by Roughness



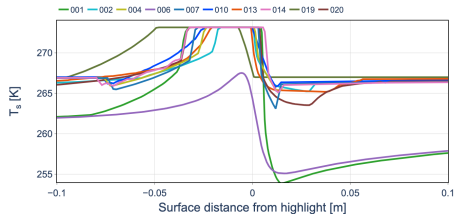
AE3933 – Grouped by Roughness

Observations

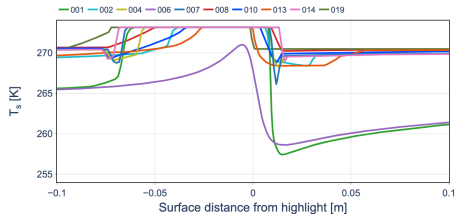
- Most submissions exhibit limited grid sensitivity in the integrated Q'_c .
- Grouping by prescribed roughness reveals a systematic roughness effect, although significant code-to-code variability remains within the roughness groups.



NACA0012 – Surface Temperature Comparison (L1)



AE3932

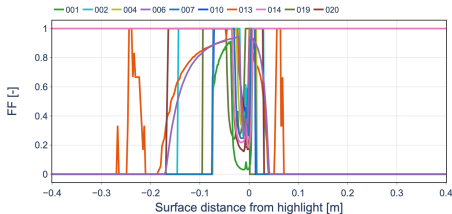


AE3933

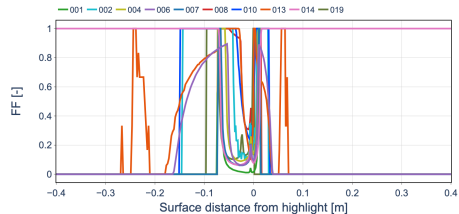
Observations

- Substantial code-to-code variability is observed in the surface-temperature distributions for both conditions.
- AE3933 exhibits a wider region where the surface temperature remains near the freezing temperature, consistent with its warmer freestream condition.
- The extent of this near-freezing region directly influences the predicted freezing-fraction distribution and, ultimately, the ice-accretion limits.

NACA0012 – Freezing Fraction Comparison (L1)



AE3932



AE3933

Observations

- Considerable code-to-code variability is observed in the freezing-fraction distributions for both conditions.
- AE3932 generally exhibits a narrower low-freezing-fraction region near the leading edge, whereas the warmer AE3933 condition tends to produce a wider region.
- This trend is consistent with the broader near-freezing surface-temperature region observed for AE3933.
- However, the large code-to-code variability and differences in transition locations make the freezing-fraction distributions difficult to interpret quantitatively.

NACA0012 – HTC & Thermodynamics Assessment

Convective Heat Transfer

- Local HTC exhibits substantial code-to-code variability, with prescribed roughness explaining only part of the spread.
- Reconstructed T_{rec} is closely clustered, consistent with the agreement previously observed in C_p .
- Despite the local HTC differences, integrated Q'_c exhibits limited grid sensitivity for most submissions.

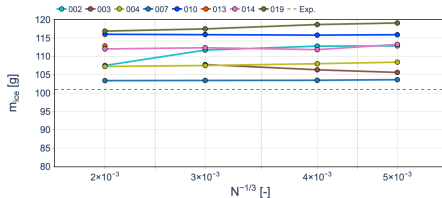
Thermodynamic Response

- Surface temperature and freezing fraction exhibit substantially greater code-to-code variability.
- The warmer AE3933 condition generally produces a wider near-freezing surface-temperature region and a correspondingly wider low-freezing-fraction region.
- The large variability in freezing fraction and transition locations limits more quantitative code-to-code interpretation.

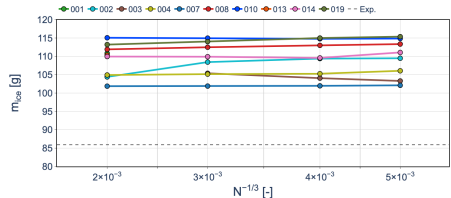
Takeaway

Convective heat transfer is relatively insensitive to grid refinement but exhibits substantial code-to-code variability. The warmer AE3933 condition produces a broader near-freezing region, which is expected to influence the freezing behavior and subsequent ice-accretion limits.

NACA0012 – Ice Mass Grid Convergence (15 Bins)



AE3932



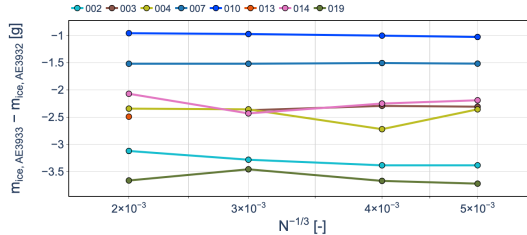
AE3933

Observations

- Most submissions exhibit limited sensitivity of the predicted ice mass to spatial grid refinement for the 15-bin distribution.
- AE3933 generally produces less ice mass than AE3932, with predicted reductions of at most approximately 3%, compared with the approximately 15% observed experimentally.



NACA0012 – Ice Mass Difference Between Conditions (15 Bins)

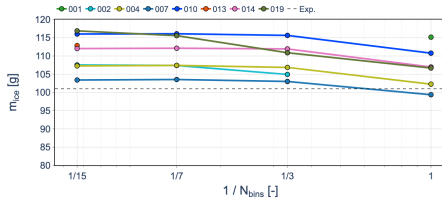


Observations

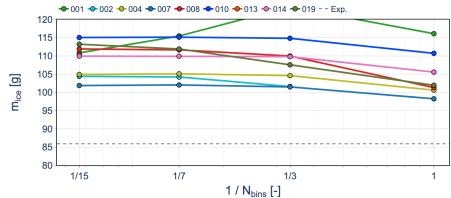
- The matched-participant comparison confirms systematically lower ice mass for AE3933 than AE3932 for most submissions.
- The numerical predictions reproduce the experimental trend of lower ice mass for AE3933, but predict a substantially smaller difference than the approximately 15% observed experimentally.



NACA0012 – Ice Mass Sensitivity to Droplet Distribution (L1)



AE3932



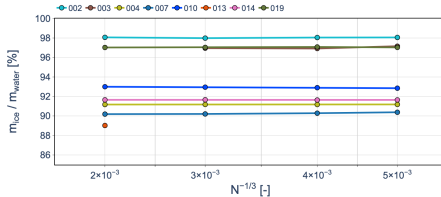
AE3933

Observations

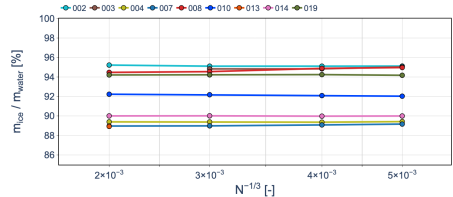
- The 15- and 7-bin distributions generally produce comparable ice masses for both conditions.
- Sensitivity increases when the distribution is reduced to 3 bins and becomes more pronounced for the single-bin representation.



NACA0012 – Ice-to-Water Mass Ratio



AE3932



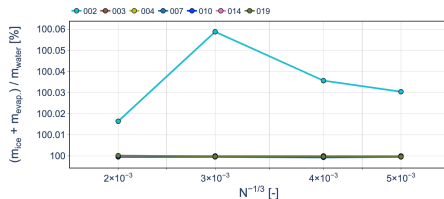
AE3933

Observations

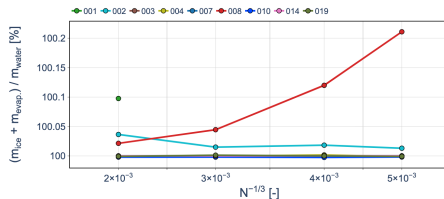
- Approximately 90% of the impinging water is converted into ice for both conditions.
- AE3933 consistently exhibits a lower ice-to-water ratio than AE3932, indicating greater water loss.
- Since the water impingement is nearly identical between the two conditions, this difference originates primarily from the thermodynamic mass balance.



NACA0012 – Ice and Evaporation-to-Water Mass Ratio



AE3932



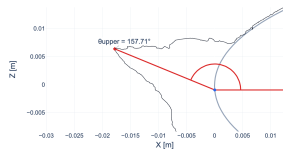
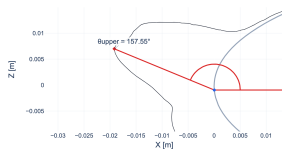
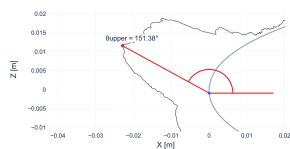
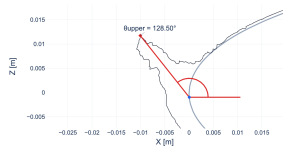
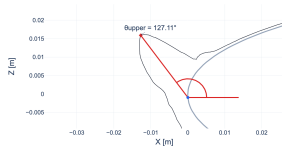
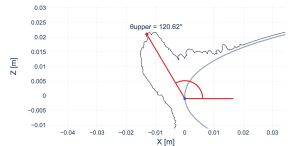
AE3933

Observations

- The combined ice and evaporated-water mass accounts for approximately 100% of the impinged water for most submissions.
- This indicates that the lower ice-to-water ratio for AE3933 is primarily associated with increased evaporation.
- Values slightly above 100% are observed for a few submissions and are likely attributable to numerical integration or round-off errors.



NACA0012 – Experimental Upper Horn Angle

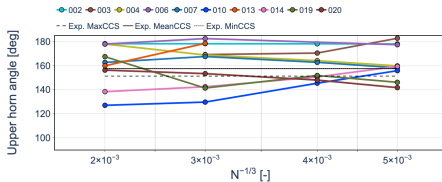
**MinCCS****MeanCCS
AE3932****MaxCCS****MinCCS****MeanCCS
AE3933****MaxCCS**

Experimental Reference

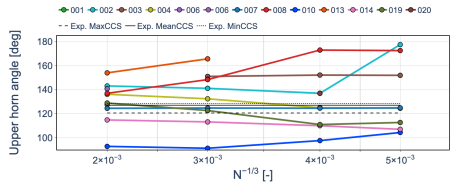
- The experimental upper-horn angle ranges from approximately $151\text{--}158^\circ$ for AE3932 and $121\text{--}129^\circ$ for AE3933.



NACA0012 – Upper Horn Angle Grid Convergence (15 Bins)



AE3932

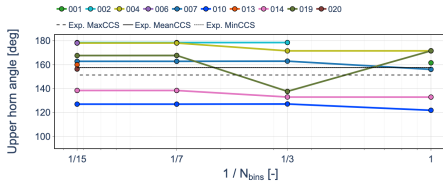


AE3933

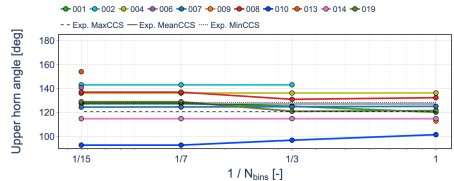
Observations

- Horn-angle predictions exhibit substantial code-to-code variability and do not show systematic grid convergence across the participant ensemble.
- AE3932 becomes more tightly clustered on the coarser grids, whereas AE3933 exhibits increasing spread with grid coarsening.
- A consistently lower upper-horn angle is observed for AE3933 compared with AE3932, in agreement with the experimental trend.
- At L1, the code-to-code scatter remains comparable between conditions, with relative IQRs of approximately 16% for AE3932 and 13% for AE3933.

NACA0012 – Upper Horn Angle Sensitivity to Droplet Distribution (L1)



AE3932



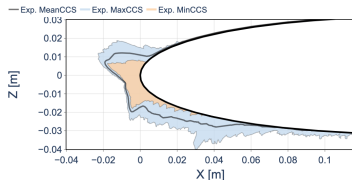
AE3933

Observations

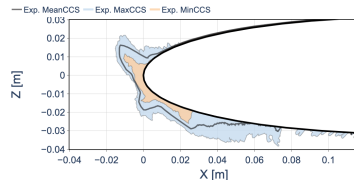
- The upper-horn angle exhibits only small variations across the different droplet-size distributions for most submissions.
- The overall horn orientation is therefore relatively insensitive to the droplet-size discretization.



NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932 – Experiment

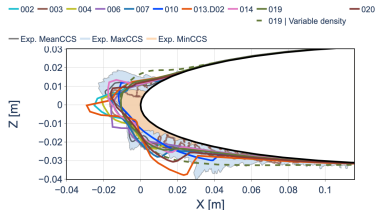


AE3933 – Experiment

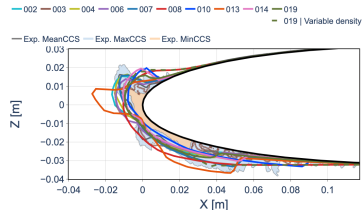
Observations

- The experimental ice shapes show a clear geometric distinction between AE3932 and AE3933, particularly in upper-horn orientation.
- The code-to-code variability observed in the upper-horn angle is directly reflected in the spread of the predicted ice shapes.
- The experimental condition-to-condition distinction is reproduced to varying degrees by the numerical predictions.
- Grouping by prescribed roughness reveals some systematic roughness dependence.

NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932

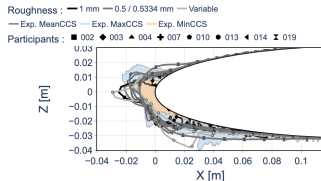


AE3933

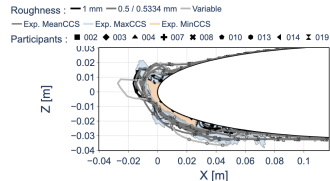
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NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932 – Grouped by Roughness

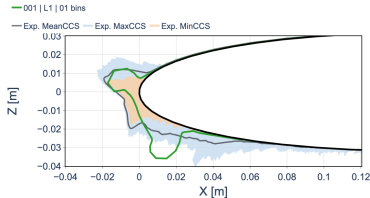


AE3933 – Grouped by Roughness

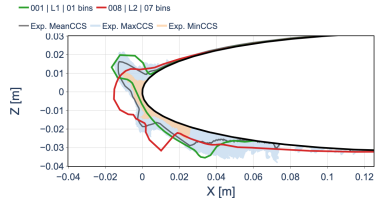
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- The experimental condition-to-condition distinction is reproduced to varying degrees by the numerical predictions.
- Grouping by prescribed roughness reveals some systematic roughness dependence.

NACA0012 – Multilayer Ice-Shape Comparison (L1)



AE3932



AE3933

Observations

- The available multilayer predictions suggest improved agreement with the experimental ice shapes and capture more of the observed geometric features than the single-layer results.
- The limited number of multilayer submissions, however, prevents drawing a conclusive assessment of this improvement across the participant ensemble.



NACA0012 – Ice Accretion Summary

Verification

- Ice mass shows limited grid sensitivity, while horn orientation exhibits greater code-to-code variability; 15- and 7-bin distributions generally provide comparable results.

Condition-to-Condition Response

- AE3933 generally produces less ice than AE3932, although the predicted reduction remains substantially smaller than the approximately 15% observed experimentally.
- A lower upper-horn angle is consistently predicted for AE3933, reproducing the experimentally observed geometric shift between the two conditions.

Ice Geometry

- Considerable code-to-code variability remains in the predicted ice shapes, particularly in horn orientation and local geometry.
- The available multilayer predictions suggest improved agreement with experiment and capture additional geometric features, although the limited number of submissions prevents a conclusive assessment.

Takeaway

The main distinction between AE3932 and AE3933 emerges in the thermodynamic response and resulting ice accretion, rather than in the upstream aerodynamics or water impingement.

NACA0012 – Overall Assessment

Aerodynamics

- Overall, the aerodynamic predictions are well clustered; C_p agrees well with experiment despite differences in prescribed roughness, while C_D exhibits the greatest sensitivity to spatial refinement.

Droplet Impingement

- Predictions remain relatively consistent across the grid hierarchy, with small grid-to-grid variations, and 15- and 7-bin distributions provide comparable results.

Convective Heat Transfer & Thermodynamics

- HTC exhibits substantial code-to-code variability, while the warmer AE3933 condition produces the expected broader near-freezing surface-temperature region.

Ice Accretion

- AE3933 consistently produces less ice than AE3932, although the predicted reduction remains well below the approximately 15% observed experimentally.
- A lower upper-horn angle is consistently predicted for AE3933, reproducing the experimentally observed geometric shift between the two conditions.
- Considerable code-to-code variability remains in the detailed ice geometry, while the available multilayer predictions suggest improved capture of experimental geometric features.